Full-Waveform Inversion for Ring-Array Ultrasound: Algorithm, Simulation Setup, and Abdominal CT Results

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Abstract

This note describes the time-domain full-waveform inversion (FWI) implemented in `invert_ring.py`, which recovers a two-dimensional sound-speed map from ring-array ultrasound traces, and `totalseg_to_speed.py`, which builds abdominal sound-speed maps from an open, expert-segmented CT dataset and a table of tissue acoustic properties. On a TotalSegmentator abdomen (1450–3750 m/s, 128 elements), three-stage FWI (10–50, 50–125 and 100–250 kHz on 4.0, 1.83 and 1.5 mm grids, with total-variation regularization), started from uniform water, recovers the organs, fat, muscle and bone: the interior RMS error falls from 293 to 84 m/s and the soft-tissue RMS error from 53 to 14 m/s (4 m/s more than 10 mm from bone; liver 2.5, kidney 4.4 m/s), and the median bone speed reaches 3785 against a true 3750 m/s. Two implementation bugs had masked this: the multiscale handoff rescaled the model by array shape although the CT stages cover different fields of view, and batched runs never excluded the transmitter's own channel. The note records both and the runs that exposed them.

1 Forward model

The forward model is the 2D constant-density acoustic wave equation, discretized with second-order finite differences in space and time (FDTD). Sound speed $c(\mathbf{x})$ is the only material property; density and attenuation are constant:

$$\frac{\partial^2 u}{\partial t^2} = c(\mathbf{x})^2 \, \nabla^2 u + s(t) \, \delta(\mathbf{x} - \mathbf{x}_{\mathrm{tx}}). \tag{1}$$

- **Time stepping.** A leapfrog scheme advances the pressure, $u^{n+1} = 2u^n - u^{n-1} + \Delta t^2 c^2 L[u^n]$, where L is the 5-point Laplacian applied as a 3×3 convolution.
- **Stability.** The step obeys the 2D CFL bound $c_{\max} \, \Delta t \, \sqrt{1/\Delta x^2 + 1/\Delta y^2} < 1$. The code uses a CFL number of 0.45, with $c_{\max}$ the larger of the true medium maximum and the upper reconstruction bound.
- **Boundaries.** First-order Mur absorbing conditions on all four edges; corners average the x- and y-direction updates.
- **Sources and receivers.** A transmitter is a soft (additive) point source at its nearest grid node. Receivers sample pressure by bilinear interpolation, so element positions are not locked to pixel centres.
- **Execution.** Fields stay on one PyTorch device (CPU, CUDA or Apple MPS). Independent transmitter shots are advanced together as a batch, and on CUDA the step is compiled with `torch.compile`.

2 Simulation setup

A ring of point transducers surrounds the object; each element both transmits and receives. One element fires a chirp per shot and every element records, giving an $N \times N \times$ time dataset. Two configurations are used: the default phantom setup (200 mm field of view, 70 mm ring) and the abdominal setup, which scales the grid and ring to fit a 0.3 m-wide torso.

2.1 Acquisition and discretization

Table 1: Simulation parameters. Abdomen values (TotalSegmentator subject s1366) are listed per multiscale stage.

Parameter	Phantom setup (default)	Abdomen setup, per stage
Grid (cells)	401×401	$101^2 \rightarrow 201^2 \rightarrow 401^2$
Grid spacing Δx	0.50 mm	4.00 $\rightarrow$ 1.83 $\rightarrow$ 1.50 mm
Field of view	200 mm	400 $\rightarrow$ 366 $\rightarrow$ 600 mm
Ring radius	70 mm	167.7 $\rightarrow$ 168.3 $\rightarrow$ 168.5 mm
Elements	256 (pitch 1.7 mm)	128 (pitch 8.3 mm)
Coupling medium	water, 1500 m/s	water, 1480 m/s
Source pulse	Hann-windowed linear chirp, 100–250 kHz, 20 μs	10–50 kHz / 80 μs $\rightarrow$ 50–125 kHz / 40 μs $\rightarrow$ 100–250 kHz / 20 μs
Time step Δt (CFL 0.45)	$\approx$ 80 ns	326 $\rightarrow$ 149 $\rightarrow$ 122 ns
Record length	1.6$\times$ ring diameter / slowest speed	$\approx$ 371 μs (1135 $\rightarrow$ 2490 $\rightarrow$ 3039 steps)
Points per shortest wavelength	≈ 12	$\approx 7.3 \rightarrow 6.3 \rightarrow 3.9$
Inversion bounds on c	1400–2000 m/s	1400–3900 m/s

The abdomen time steps follow from the 3900 m/s upper bound, which sets the CFL limit. The record length is 1.6 times the time to cross the ring diameter at the slowest speed in the medium (fat, 1450 m/s). The grid is never finer than the CT pixels, so the final stage runs at the 1.5 mm CT spacing on a 600 mm field of view, most of it coupling water. Because each stage's spacing comes from the CT, the three stages cover different fields of view.

2.2 Abdominal sound-speed model from an open segmentation

The medium is one axial slice of a public, expert-segmented CT, converted by `totalseg_to_speed.py`. Organs, vessels, bowel, bone and the deep back muscles come from the segmentation; only pixels no mask covers are classified by Hounsfield units (HU).

1. **Data.** The *Small subset of TotalSegmentator Dataset* v2.0.1 [5, 4] holds 102 CT volumes resampled to 1.5 mm, each with masks of 117 structures (CC BY 4.0). `-fetch` downloads a single subject out of the 3.2 GB zip with HTTP range requests (44 MB for subject s1366).

2. **Subject choice.** Of the 54 subjects whose masks include the main upper-abdominal organs, the cleanest were screened by compressed bytes per voxel, then by the HU standard deviation inside the liver after removing a 5^3-voxel local mean. Subject s1366 has 14.7 HU of noise, against 140 HU for the noisiest (low-dose) scan, s0407, which was rejected.

3. **Slice choice.** Among slices showing all ten key structures (liver, spleen, both kidneys, pancreas, stomach, aorta, inferior vena cava, both deep back muscles), the one with the least gas inside the body is used: slice 259, at the T12/L1 level, with 2.5% gas.

4. **Tissue classes.** Masks are merged into the classes of Table 3. Unlabelled body pixels are classified by HU after a 5×5 median filter: fat (-190 to -30), gas (< -190) or other soft tissue. Gas, which the scalar fluid model cannot represent, becomes water; inside segmented solid organs low HU is treated as noise, not gas. The body is the largest connected region above -500 HU, which excludes the arms and the scanner table.

5. **Acoustic properties per class.** Each class points to one row of `abdominal_acoustic_parameters.csv` (Table 2), compiled from [1, 2, 3]. The script writes speed, density, power-law attenuation and B/A maps; the present solvers are lossless and constant-density, so only speed enters them. Gas is modelled as water rather than 343 m/s air: without a density contrast the scalar model cannot reproduce the near-total reflection at a gas boundary.

6. **Embedding.** `load_ct_ring_grid` centres the body, resamples it to each stage's grid with nearest-neighbour sampling, pads the outside with 1480 m/s water, and places the ring 10 mm outside the body with an 8-cell margin to the grid edge.

Table 2: Tissue acoustic properties (`abdominal_acoustic_parameters.csv`; values compiled from [1, 2, 3]). Attenuation follows $\alpha = \alpha_0 f^{\,y}$ with f in MHz.

Tissue	c (m/s)	ρ (kg/m^3)	α_0 (dB/MHz^y/cm)	y	B/A
Soft tissue (machine average)	1540	1000	0.5	1.1	7.0
Water / bile / urine	1480	1000	0.0022	2.0	5.0
Fat (visceral/subcutaneous)	1450	920	0.75	1.25	10.0
Liver	1570	1060	0.6	1.2	6.8
Kidney	1560	1050	0.9	1.0	7.2
Blood	1575	1050	0.15	1.3	6.1
Muscle (abdominal wall)	1580	1080	1.1	1.0	7.4
Spleen	1567	1054	0.55	1.2	6.8
Bowel gas / air	343	1.2	12.0	2.0	0.4
Bone / cortical rib	3750	1900	15.0	1.5	0.0

Table 3: Tissue classes of the s1366 slice (333×333 pixels at 1.5 mm) and the table row each uses.

Class	Source	Property row	c (m/s)	Share of body
Fat	HU	Fat	1450	31.9%
Other soft tissue	HU	Soft tissue average	1540	23.5%
Liver	mask	Liver	1570	21.8%
Digestive tract (stomach, small bowel, colon)	masks	Soft tissue average	1540	7.1%
Muscle (deep back muscles)	masks	Muscle	1580	4.3%
Gas, modelled as water	HU	Water	1480	2.5%
Bone (T11–L1 vertebrae, ribs)	masks	Bone	3750	2.3%
Kidney	masks	Kidney	1560	2.1%
Blood vessels (aorta, IVC, portal/splenic vein)	masks	Blood	1575	1.9%
Other organs (pancreas, adrenals)	masks	Soft tissue average	1540	1.8%
Spleen	mask	Spleen	1567	0.8%

The map spans 1450–3750 m/s. Soft-tissue contrasts are small (within 7% of water), while bone is 2.5 times faster than water: it dominates both the data and the difficulty of the inversion.

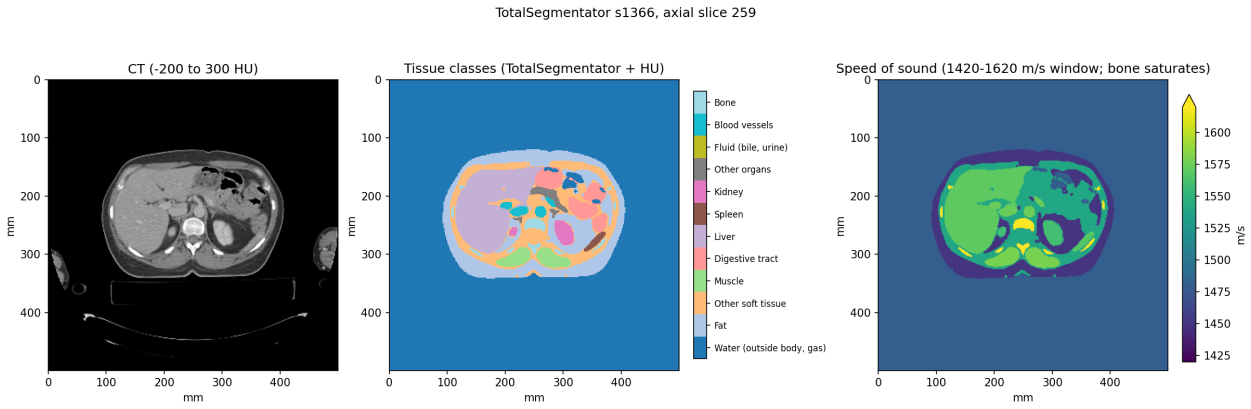


Figure 1: TotalSegmentator subject s1366, axial slice 259: CT (window −200 to 300 HU), merged tissue classes, and the resulting sound-speed map. The speed map is shown in a 1420–1620 m/s soft-tissue window; bone (3750 m/s) saturates at the top of the scale.

Earlier model. The first experiments (Section 6.3) used `ct_to_speed.py` on one slice of C4KC-KiTS case KiTS-00136 (512 × 512 at 0.684 mm). That pipeline has no anatomical labels: it thresholds HU into air, fat, soft tissue and bone, and splits superficial from deep tissue at 25 mm depth (fat into subcutaneous and visceral, soft tissue into muscle and organs), with representative speeds from Table 1 of [6] (1478–3476 m/s). The segmentation-based model replaces that depth rule with real organ, vessel and muscle boundaries.

3 Inversion algorithm

The inversion minimizes the least-squares misfit between simulated and observed traces over slowness squared $m = 1/c^2$, using an exact discrete adjoint for the gradient and first-order updates. Everything below is implemented in `invert_ring.py` and `fdtd2d/inversion/`.

3.1 Unknowns and constraints

- **Parameter.** $m = 1/c^2$ on every pixel inside an interior disk Ω of radius (ring radius − margin), with a 2-pixel margin by default. Outside Ω, c is fixed at the coupling-medium speed. The wave operator is linear in $c^2 = 1/m$, which keeps the gradient simple.
- **Bounds.** After every update m is projected onto $[1/c_{\max}^2,\ 1/c_{\min}^2]$ and the exterior is reset to the background.
- **Starting model.** Homogeneous coupling medium (1500 or 1480 m/s) inside the mask. For diagnosis only, `--start-blurred-truth` σ starts from the true slowness blurred by a Gaussian of σ mm.

3.2 Objective

$$J(m) = \frac{1}{2} \sum_{s \in S} \sum_{r \neq s} \sum_{n} \left(d_{s,r,n}^{\text{pred}}(m) - d_{s,r,n}^{\text{obs}} \right)^2 + \frac{\alpha}{2} \left\| \nabla m \right\|_{\Omega}^2 + \beta \sum_{\Omega} \left(\sqrt{|\nabla m|^2 + \varepsilon^2} - \varepsilon \right). \qquad (2)$$

S is the set of active transmitters, r runs over receivers and n over time samples. The transmitter's own channel is excluded because a point source's direct pressure is a poorly conditioned datum. (In

batched runs before commit `d519e92` this exclusion silently failed; see Section 6.1.) The Tikhonov term penalizes squared differences between neighbouring pixels inside Ω, with weight α (`--reg`). The total-variation (TV) term penalizes the magnitude of the forward-difference gradient, smoothed by ε (`--tv-eps`, default 10^{-9} in units of m, about 2 m/s at 1500 m/s), with weight β (`--tv`). Because TV grows linearly with a jump, it suppresses speckle while leaving sharp organ and bone boundaries much cheaper than the quadratic Tikhonov term does. Both weights default to 0 and can be set per stage.

Observed data are synthetic: at the start of each stage the same solver simulates all N transmitters through the true medium, without noise.

3.3 Adjoint-state gradient

For each active shot the forward pass stores the full wavefield history u^n. The adjoint field λ is then stepped backward in time with the transposed operator $L(c^2\,\cdot)$, driven by the residuals scattered onto the grid by the transpose of the bilinear receiver sampling. The gradient with respect to c^2 accumulates as

$$g_{c^2} = \Delta t^2 \sum_n \lambda^n \odot L\big[u^{n-1}\big], \qquad \frac{\partial J}{\partial m} = -\frac{g_{c^2}}{m^2} + \nabla_m R(m), \tag{3}$$

where R is the sum of the Tikhonov and TV terms. The gradient is set to zero outside the mask. With `--store-every` k the sum runs over every k-th forward step, weighted by k, and `--wave-dtype float16` stores the forward fields at half precision; on the test problem these change the gradient by 2×10^{-7} ($k = 4$) and 9×10^{-5} (float16). The adjoint reuses the forward Mur boundary update, which is not its exact transpose; the finite-difference gradient check (Section 5) bounds the effect of this.

3.4 Optimizers

Table 4: Optimizers available through `--optimizer`.

`--optimizer`	Update	Notes
`adam` (default)	Projected Adam on m, $\beta_1 = 0.9$, $\beta_2 = 0.999$	Step size `--adam-lr` in units of m ($\mathrm{s}^2/\mathrm{m}^2$), per stage: 10^{-9} default; the abdomen run uses 10^{-8} on the coarse stage and 4×10^{-9} on stages 2 and 3. Each pixel moves by at most about the step size per iteration. Tolerates the noisy gradients of shot sub-sampling.
`cg`	Polak–Ribière nonlinear CG, Armijo backtracking, restart when $\beta < 0$	Deterministic; conjugacy is lost when shots are re-sampled each evaluation.
`gradient-descent`	Projected fixed step `--step-size`	Baseline.

`--model neural` replaces the per-pixel unknowns with the weights of a Fourier-feature coordinate network; the same adjoint gradient is passed through the network by the chain rule.

3.5 Shot sub-sampling

Each gradient evaluation uses K transmitters drawn at random without replacement (`--n-shots`, default 4; `--shot-seed` fixes the sequence), so one evaluation costs $2K$ wave solves instead of $2N$.

Shots run in GPU batches of up to `--parallel-shots`. Its default, `auto`, sizes each stage's batches from free GPU memory and splits the shots into equal batches; the abdomen run fits all 32 shots in one batch.

3.6 Multiscale continuation

`--multiscale` runs coarse-to-fine stages on progressively finer grids (Table 1). Each stage regenerates its own observed data, and the final m of one stage is transferred to the next by bicubic interpolation in physical coordinates: the new grid samples the old model at its own cell centres, with background outside the old field of view. (Resizing by array shape alone, as the code first did, is correct only when every stage covers the same field of view; Section 6.2.) `--stages` sets how many stages run (default 3). `--coarse-band` sets the first stage's chirp band (default 25–75 kHz; the abdomen uses 10–50 kHz). Starting at a low frequency lengthens the period relative to the initial travel-time errors, which reduces cycle skipping: fitting a trace one period off, which traps inversions in a wrong model. `--n-elements`, `--max-iter`, `--adam-lr`, `--reg` and `--tv` take either one value or one value per stage, and `--save-model X.npz` also saves every earlier stage as `X_stage`k`.npz`.

4 Algorithm at a glance

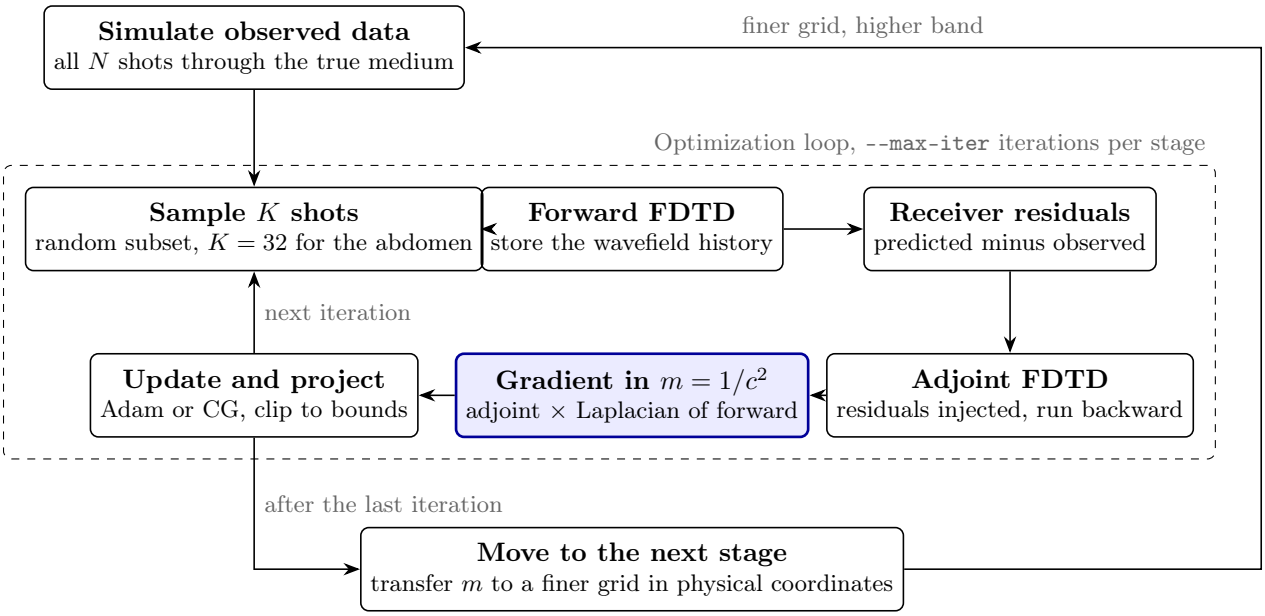


Figure 2: FWI loop inside each multiscale stage. Every iteration costs one forward and one adjoint solve per sampled shot; in the abdomen run the three stages repeat the whole loop at 10–50, 50–125 and 100–250 kHz.

5 Implementation and verification

The adjoint gradient matches finite differences to second order, and the cost of one iteration is two wave solves per sampled shot plus the stored wavefield.

- **Batching.** `TorchFDTD2DBatch` advances independent shots as one (batch, n_y, n_x) tensor. Each forward step is fused with the receiver gather, and each adjoint step with the residual injection and the gradient accumulation, so a time step is one compiled call (`torch.compile(mode="reduce-overhead")` on CUDA). Against the earlier unfused code this is about 9×, 6× and 2× faster per shot on the three abdomen grids.
- **Memory.** The forward pass keeps the wavefield of every shot in the batch for the adjoint: $n_{\text{steps}} \times n_y \times n_x \times 4$ bytes per shot in float32 (1.95 GB on a 401^2, 3039-step grid). Storing every fourth step in float16 cuts this 8×, so all 32 shots of the abdomen run fit in one batch on every stage (about 10 GB peak).
- **Gradient check.** `python invert_ring.py --gradient-check --dtype float64` compares the adjoint directional derivative $g \cdot \delta m$ with central differences on a 41×41 grid with a 12 mm, 1800 m/s inclusion (Table 5).

Table 5: Adjoint gradient check (float64, run on September 28, 2026).

Step ε	Relative error
10^{-2}	3.5×10^{-3}
10^{-3}	3.4×10^{-5}
10^{-4}	3.4×10^{-7}

The error falls 100× per 10× smaller step, the expected behaviour for a correct gradient with a second-order difference; the approximate adjoint of the Mur boundary does not show at this level. A pytest suite (`tests/`) checks the neural parameterization's chain rule the same way on ring and arc problems, and checks that the fused batched gradient matches an unbatched reference to 10^{-12}, the subsampled and float16 gradients, and the TV gradient against finite differences, and that the stage transfer places a disk correctly between grids with different fields of view.

6 Abdomen imaging results

Three-stage FWI of the segmentation-based s1366 abdomen, started from uniform 1480 m/s water, recovers soft tissue and bone: the interior RMS error falls from 293.1 to 84.2 m/s and the soft-tissue RMS error from 53.3 to 13.6 m/s. Section 6.2 records the runs that led to this setup, including the two bugs that had hidden it.

6.1 Segmentation-based abdomen (TotalSegmentator s1366)

The run used the s1366 map of Section 2.2 and the setup of Table 1, on an NVIDIA RTX A6000 (desmond):

```
python invert_ring.py --device cuda --compile on --no-show \
  --ct-speed totalseg_s1366_acoustic.npz \
  --multiscale --stages 3 --coarse-band 10 50 --n-elements 128 \
  --n-shots 32 --max-iter 200 200 100 --adam-lr 1e-8 4e-9 4e-9 \
  --reg 0 --tv 0 1e7 1e7 --store-every 4 --wave-dtype float16 \
  --c-min 1400 --c-max 3900 --save-model totalseg_s1366_s3full_model.npz \
  --save-figure totalseg_s1366_s3full_multiscale.png \
  --display-range 1420 1620 --display-diff 300 --crop-to-ring
```

The coarse stage is unregularized and uses the larger step; stages 2 and 3 add TV with $\beta = 10^7$. The log is `totalseg_s1366_s3full_log.txt`; the model after each stage is saved as `totalseg_s1366_s3full_model.npz` (stage 3), `_stage1.npz` and `_stage2.npz`.

Table 6: Final s1366 run per stage (32 random shots per iteration, one GPU batch). J at iteration 1 is after the first update; RMS errors are scored on each stage's own grid. The GPU was shared with up to five other jobs throughout, so the measured times are inflated; the idle estimates use the per-iteration times of earlier runs with the same grids and shot count.

Stage	Iter.	J, iter. 1 $\to$ last	RMS (m/s)	Measured (s)	Idle est. (s)
101^2, 4.00 mm, 10–50 kHz	200	$1.46 \times 10^5 \to 3.9 \times 10^1$	194.5	170	76
201^2, 1.83 mm, 50–125 kHz	200	$1.92 \times 10^4 \to 1.76 \times 10^3$	120.2	570	311
401^2, 1.50 mm, 100–250 kHz	100	$1.86 \times 10^4 \to 2.42 \times 10^3$	84.2	1516	630
Total				2256	≈ 1020

Table 7: Interior speed error by tissue for the final s1366 run, on the 401^2 grid. "Start" is the uniform 1480 m/s starting model; "Stage 2" is the stage-2 model transferred to the 401^2 grid before any stage-3 update.

Region	Pixels	Start	Stage 2	Final RMS	Final bias
All interior	38 249	293.1	139.9	84.2	−5
All soft tissue ($c < 1700$ m/s)	37 632	53.3	49.3	13.6	0
more than 10 mm from bone	34 287		10.1	4.1	0
within 10 mm of bone	3 345		162.3	43.7	+6
Fat	8 480	30.0	14.5	5.0	+2
Soft-tissue average (other, bowel, pancreas)	8 628	60.0	101.1	27.6	+1
Muscle	1 146	100.0	10.2	8.4	−4
Spleen	204	87.0	25.3	5.7	−5
Kidney	552	80.0	11.2	4.4	−1
Liver	5 803	90.0	13.7	2.5	−1
Blood vessels	519	95.0	8.2	6.1	−4
Water, bile, urine (incl. coupling water)	12 300	0.0	4.0	1.9	0
Bone	617	2270.0	1031.6	654.6	−306
Bone, median speed (true 3750 m/s)		1480	3034	3785	

- **Soft tissue.** Every organ is within a few m/s of its true speed: liver 2.5, kidney 4.4, fat 5.0, spleen 5.7 m/s RMS. Away from bone the difference map is blank apart from faint lines on tissue boundaries (Figure 3).
- **Bone.** The vertebral body, spinous process and ribs are in place, and the median bone speed reaches its true value. The remaining bone error is at the edges of the vertebral body and ribs, where single pixels are over- or under-estimated.
- **Near bone.** The tissue within 10 mm of bone still carries most of the soft-tissue error (43.7 m/s RMS, down from 162 m/s after stage 2), mainly the tissue enclosed by the vertebral body.
- **What stage 3 adds.** Scored on the same 401^2 grid, stage 3 lowers the interior error from 139.9 to 84.2 m/s, the near-bone error from 162 to 44 m/s and lifts the median bone speed from 3034 to 3785 m/s.

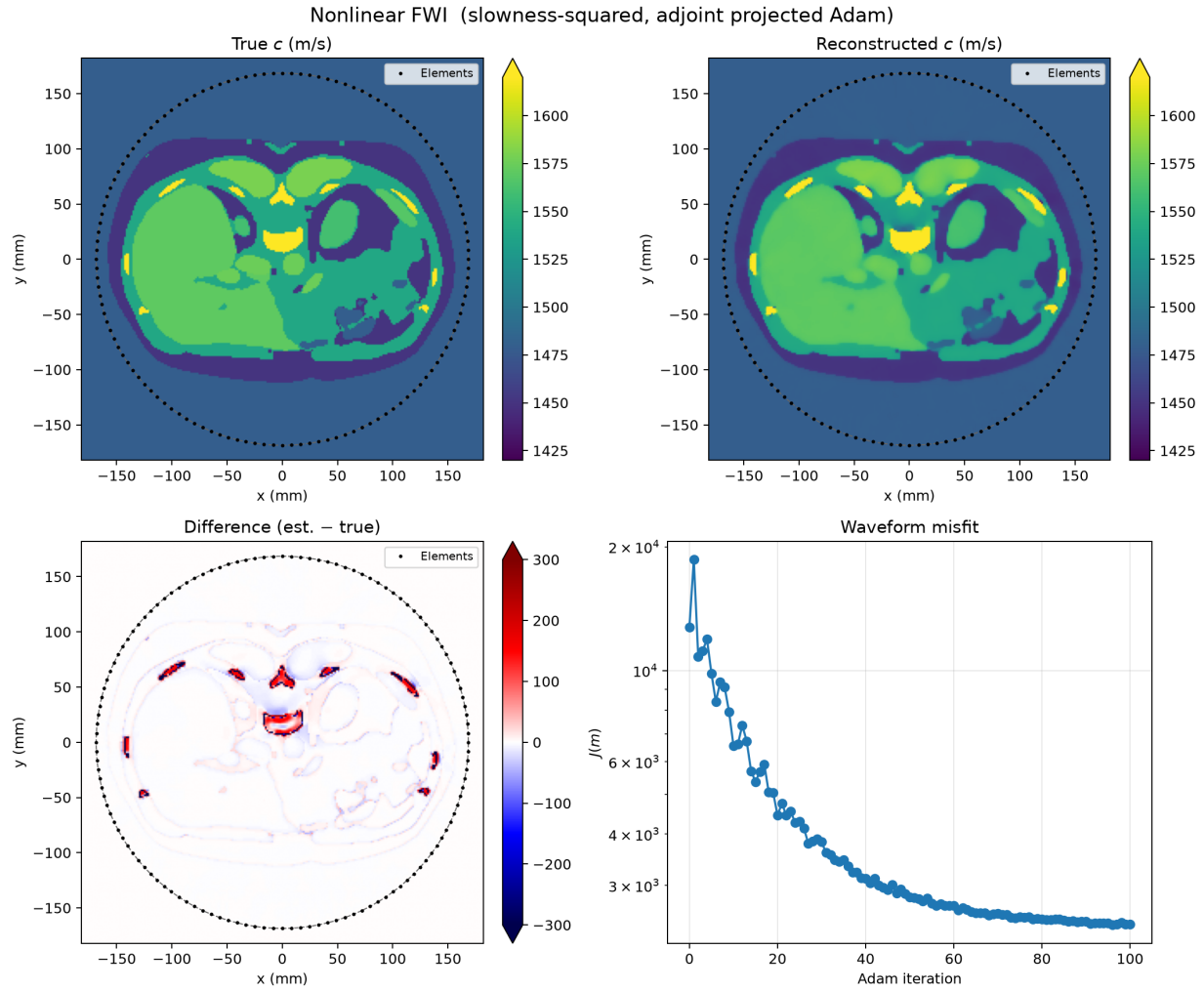


Figure 3: Three-stage FWI of the s1366 abdomen, cropped to the ring. True c and the final reconstruction in a 1420–1620 m/s window (bone saturates), difference clipped at ±300 m/s, and the stage-3 misfit. The plot's y axis runs anterior (bottom) to posterior (top).

- **Convergence.** The coarse stage fits its data almost exactly. Stage 3's J falls 8× and is nearly level by iteration 100 (4% lower than at iteration 75).

6.2 How the setup was reached

Table 8 lists the s1366 runs in order. All started from uniform water unless noted. Until the stage-transfer fix (last three rows), every three-stage run handed stage 3 a magnified model; the errors are those of each run's output on its own final grid.

- **Bug 1: transmitter channel.** Batched runs never excluded the transmitter's own channel from the misfit: the zeroing acted on a copy made by advanced indexing. It affects the first two rows and the KiTS run (Section 6.3), and was found by a test comparing the batched gradient with an unbatched reference.

- **Shots, elements and regularization.** More shots per iteration did not remove the artifacts;

Table 8: s1366 runs on the RTX A6000, 128 elements unless noted. RMS errors in m/s, interior / soft tissue ($c < 1700$ m/s). Uniform water scores 293 / 53 on the 401^2 grid and 297 / 54 on the 201^2 grid.

Run	Final grid	Error
Three stages, before either fix		
4 shots, no regularization, 25–75 kHz coarse band	401^2	315 / 204
64 shots	401^2	338 / 252
After the transmitter-channel fix (32 shots unless noted)		
Tikhonov 10^{14}	401^2	314 / 216
+ 10–50 kHz coarse band, Tikhonov 10^{15}	401^2	288 / 170
+ 256 elements on stages 2–3	401^2	304 / 195
coarse stage 200 iterations at 10^{-8}	401^2	293 / 180
Tikhonov 0 / 10^{14} / 10^{14}	401^2	353 / 268
same, started from the true map blurred by 10 mm	401^2	350 / 265
TV 0 / 10^7 / 10^7	401^2	353 / 269
two stages, 50 stage-2 iterations	201^2	185 / 32
two stages, 200 stage-2 iterations	201^2	143 / 37
two stages, 8 shots, TV 10^6	201^2	145 / 38
three stages, 8 shots, stage 3: 10 iterations at 4×10^{-10}	401^2	386 / —
After the stage-transfer fix		
same 10-iteration stage-3 test	401^2	137 / 48
its stage-2 model	201^2	119 / 27
three stages, 32 shots, stage 3: 100 iterations at 4×10^{-9} (Section 6.1)	401^2	**84 / 14**

256 elements removed the radial streaks near the ring. Tikhonov at 10^{15} suppressed speckle but charged the true bone edges about six times the converged coarse misfit, blocking the bone; TV, which charges jumps linearly, replaced it.

- **A misleading diagnostic.** Saving the model after every stage showed stages 1 and 2 working and stage 3 producing the same speckled, biased model from any input, including a 10 mm-blurred true map (Figures 4 and 5). This was first read as cycle skipping at 100–250 kHz, and stage 3 was dropped. A stage-3 test with steps too small to move any pixel more than about 7 m/s still raised the error from 145 to 386 m/s, which ruled that out.

- **Bug 2: stage transfer.** The transfer resized the model by array shape, as if every stage covered the same field of view. The s1366 stages cover ±200, ±183 and ±300 mm, so stage 2 started from a model shrunk by 8.5% and stage 3 from one magnified 1.64×, scoring 388 m/s before any update. Interpolating in physical coordinates fixed both: stage 2 improved from 145 to 119 m/s, and stage 3 now lowers the error from 140 to 84 m/s on its grid. The phantom stages share one field of view, which is why the bug did not show there. The KiTS stages (468, 425 and 406 mm) were affected too.

6.3 Earlier experiment: KiTS-00136 with the HU-threshold model

The first abdomen runs used the `ct_to_speed.py` map of C4KC-KiTS case KiTS-00136 (categorical mapping, 1478–3476 m/s). Its stages ran at 4.63, 2.12 and 1.01 mm on 468, 425 and 406 mm fields of view, with a 194.4 mm ring of 64 elements (19 mm pitch). Because those fields of view differ, this

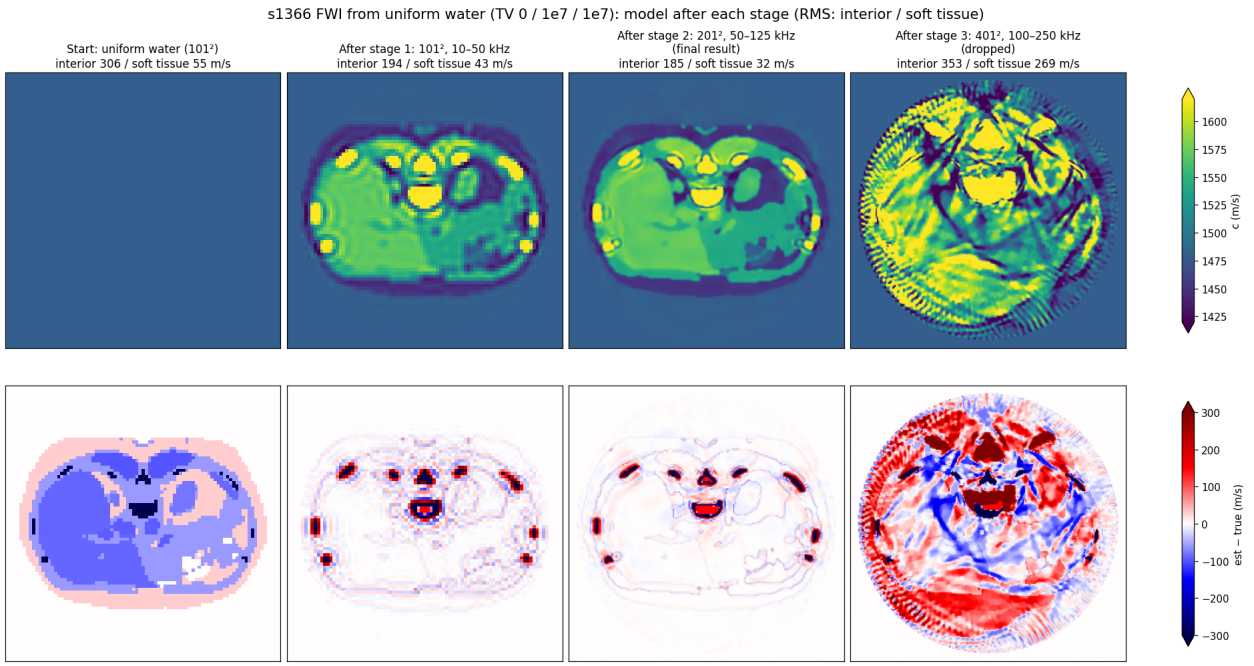


Figure 4: Before the stage-transfer fix: run with TV 0 / 10^7 / 10^7 from uniform water, speed (top) and error (bottom) at the start and after each stage. Stage 3 received the stage-2 model magnified 1.64×.

run also used the faulty stage transfer of Section 6.2, and it kept each transmitter's own channel in the misfit. Three-stage FWI recovered the soft-tissue anatomy and the outer cortex of the vertebra, but left the vertebral body hollow, underestimated by up to about 2000 m/s.

6.3.1 Full multiscale reconstruction

- **Soft tissue.** Organ boundaries, the paraspinal muscles, the abdominal-wall muscle and the fat layers appear in the right places with roughly the right speeds; the difference map is near zero across most of the abdomen.
- **Bone.** The cortical rim of the vertebral body and the posterior elements are recovered, partly at the full 3476 m/s. The interior of the vertebral body stays near the background speed, the dark region in the difference map.
- **Artifacts.** Faint radial streaks fan inward from the ring. With 64 elements the pitch is about 19 mm, several wavelengths at 100–250 kHz, so the source and receiver sampling is sparse at the finest stage.
- **Convergence.** In the final stage J falls from 5.81×10^3 to 5.78×10^2 (10×), with a rise after the first step and the noise expected from random shot sub-sampling.

The run was made on the RTX A6000 (desmond) on September 26, 2026, with

```
python invert_ring.py --device cuda --ct-speed abdomen_acoustic.npz \
  --multiscale --c-max 3600 --n-elements 64 --n-shots 16 \
  --parallel-shots 16 --adam-lr 5e-09 --max-iter 50 \
  --no-show --save-figure abdomenmultiscale.png
```

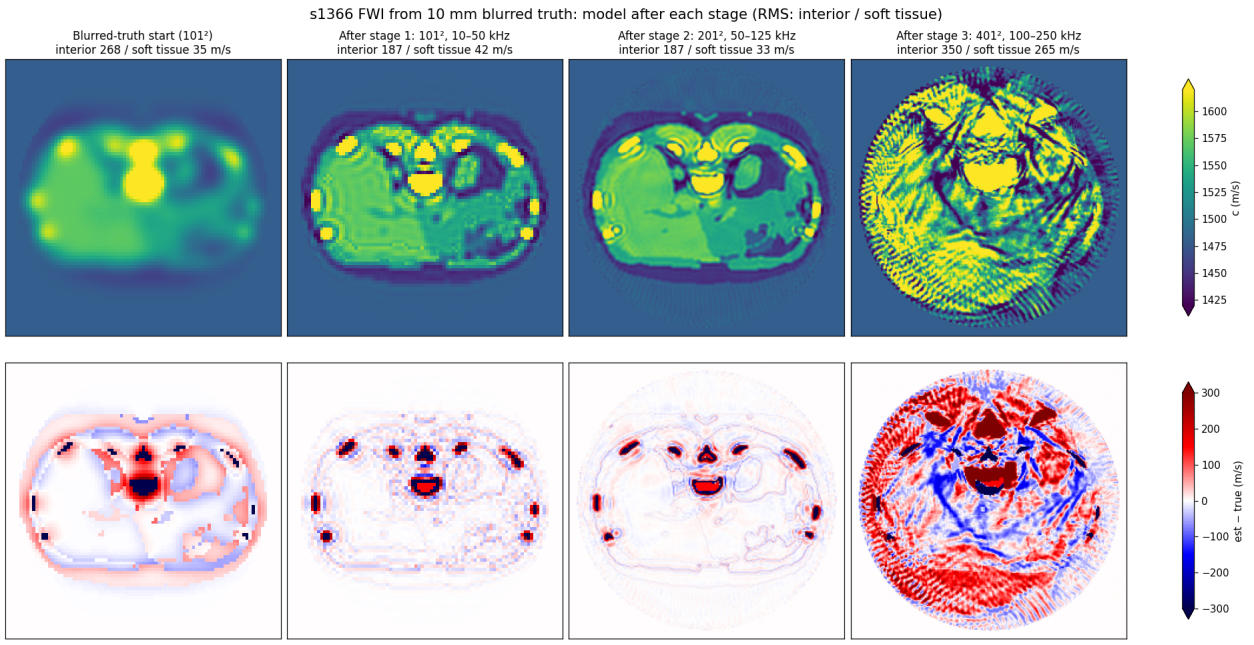


Figure 5: Before the stage-transfer fix: the same diagnostic from the true map blurred by a 10 mm Gaussian (Tikhonov 0 / 10^{14} / 10^{14}). Stages 1 and 2 converge to nearly the same model as the water start; stage 3 again starts from a magnified model.

(recovered from the shell history on desmond; the original line also carried a superseded `--save-figure abdomen_inverse.png` and a repeated `--no-show`). The output is in `abdomenmultiscale_log.txt`, and the figure on desmond is byte-identical to the committed one. All other options took the defaults of that version of the code: no regularization, $c_{\min} = 1400$ m/s, shot seed 0, float32 with a compiled step. Like every batched run before commit `d519e92`, it kept each transmitter's own channel in the misfit. Table 9 lists the logged numbers.

Table 9: KiTS-00136 multiscale run, from `abdomenmultiscale_log.txt`. J at iteration 1 is after the first Adam update.

Stage	Grid, spacing, band	J at iteration 1 $\to$ 50	Wall time
1	101^2, 4.63 mm, 25–75 kHz	$2.34 \times 10^4 \to 6.33 \times 10^3$	88 s
2	201^2, 2.12 mm, 50–125 kHz	$9.28 \times 10^3 \to 1.59 \times 10^3$	191 s
3	401^2, 1.01 mm, 100–250 kHz	$7.82 \times 10^3 \to 5.78 \times 10^2$	432 s

Elements 64; gradient shots 16 of 64 per iteration; Adam learning rate 5×10^{-9}
Final stage: 4704 steps at $\Delta t = 89.6$ ns; total wall time 711 s
Interior RMS error in c: 223.7 m/s; reconstructed range 1400–3600 m/s (true 1478–3476)

6.3.2 Coarse-stage reproduction

Before the multiscale log was recovered, the first stage was rerun on CPU on October 27, 2026 to inspect the coarse result on its own. This rerun used 4 gradient shots and a learning rate of 4×10^{-9} rather than the original 16 shots and 5×10^{-9}:

```
python invert_ring.py --ct-speed abdomen_acoustic.npz --coarse-only \
```

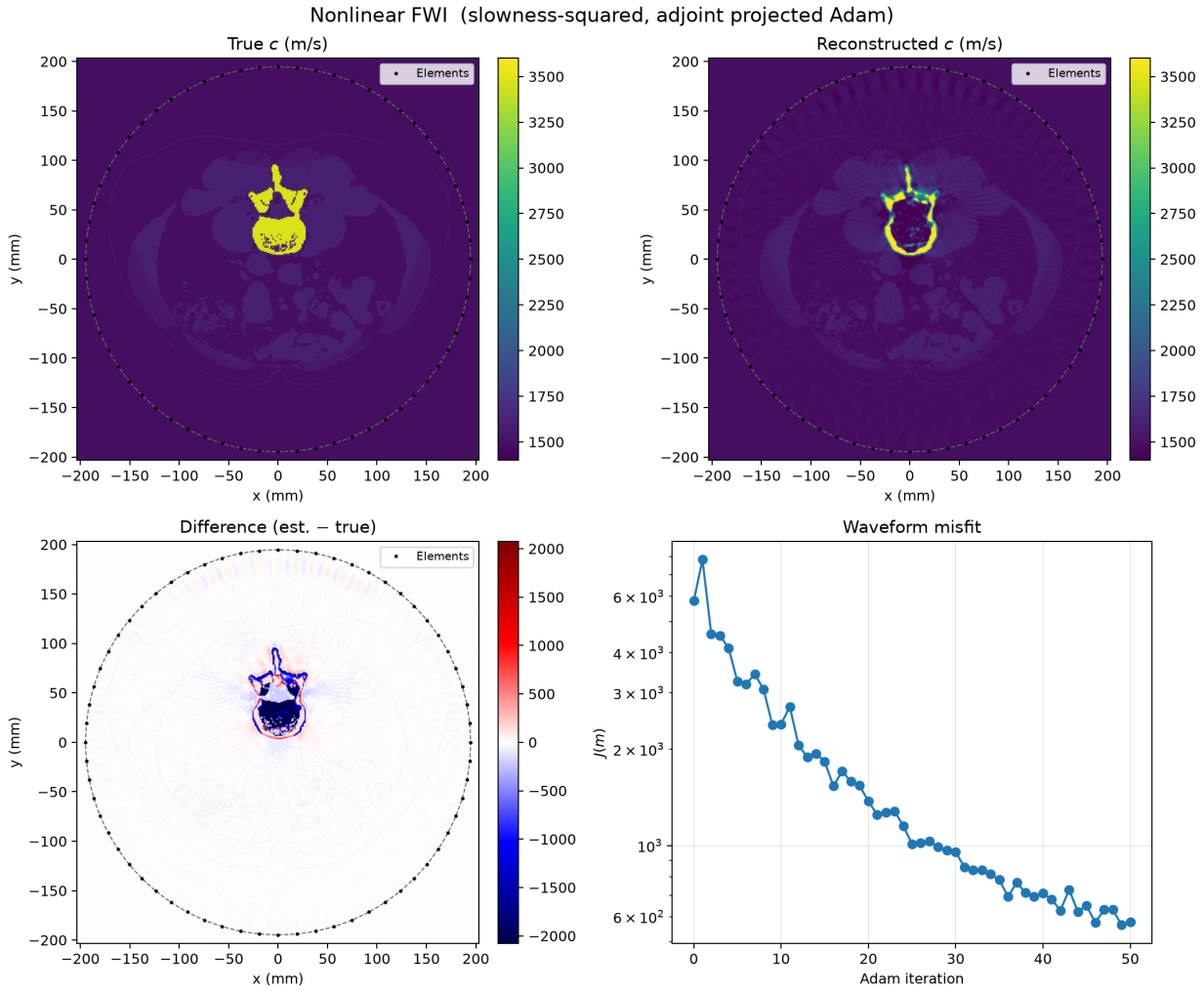


Figure 6: Multiscale FWI of the abdomen (`abdomenmultiscale.png`, committed September 26, 2026). True c, reconstruction after the 401^2 / 100–250 kHz stage, difference, and the final-stage waveform misfit over 50 Adam iterations.

```
--n-elements 64 --n-shots 4 --max-iter 50 --adam-lr 4e-9 \
--c-min 1400 --c-max 3600
```

The full log is in `abdomen_coarse_stage_log.txt`.

The soft-tissue structures appear after a single coarse stage, but the vertebra goes the wrong way: speed falls to the lower bound where it should rise to 3476 m/s. A straight path through about 40 mm of bone arrives about 15 μs earlier than through water, more than half the 13 μs period at 75 kHz, so the coarse data are already cycle-skipped there. The later stages start from this hollow vertebra and only sharpen its rim, which matches Figure 6. The RMS error is dominated by the bone pixels; the difference map is close to zero over most of the soft tissue.

Table 10: Coarse-stage abdomen reproduction.

Quantity	Value
Grid / band	101×101 at 4.63 mm, 25–75 kHz
Time steps, Δt	1029, 409 ns
Misfit J, start $\rightarrow$ end	$7.35 \times 10^3 \rightarrow 1.35 \times 10^3$ (5.4$\times$ lower)
Interior RMS error in c	298.5 m/s
Interior c range, true	1478–3476 m/s
Interior c range, reconstructed	1400–1832 m/s
Wall time (8 CPU threads, float32)	71 s (observed data 2.1 s, optimization 68.6 s)

7 Limitations and future work

- **Inverse crime.** Observed and predicted data come from the same solver on the same grid, with no noise. This is the largest caveat on the results: data should next be generated on a finer grid, or with an independent solver, with noise added.
- **Grid resolution.** The final stage has only 3.9 points per shortest wavelength (limited by the 1.5 mm CT voxels), low for a second-order scheme. Numerical dispersion is invisible here only because data and model share the grid; real data will need a fourth-order stencil or a finer grid.
- **Bone and its surroundings.** Bone reaches its true median speed, but its edges and the tissue within 10 mm of it carry most of the remaining error (43.7 m/s). A bone-aware prior, a travel-time misfit or more stage-3 iterations are the next things to try.
- **Run time.** The final run takes an estimated 17 minutes on an idle RTX A6000, 60% of it in stage 3. Fewer shots do not help on these grids (8 shots were no faster than 32); checkpointing or a custom kernel that advances several time steps per launch would.
- **Element count.** With 128 elements the pitch (8.3 mm) exceeds half a wavelength above about 90 kHz. No streaks remain in the final result, but 256 elements removed them in earlier runs and should be re-tested with the fixed transfer.
- **Physics.** Constant density, no attenuation and no shear waves. Bone and gas are only approximated by a scalar acoustic model. The density and attenuation maps written by `totalseg_to_speed.py` are not yet used by the solvers.
- **Reproducibility.** The commands and logs of all abdomen runs are saved alongside their figures. The KiTS-00136 command was recovered after the fact from shell history.

Sources

- Repository scjoshi/SoundWaveSimulation: `invert_ring.py`, `fdtd2d/torch_backend.py`, `fdtd2d/inversion/` (CG, misfit, medium), `fdtd2d/ct_medium.py`, `ct_to_speed.py`, `fdtd2d/inversion/neural.py`.
- Final s1366 run: `totalseg_s1366_s3full_multiscale.png`, `totalseg_s1366_s3full_log.txt` and the saved models; earlier runs in Table 8 as `totalseg_s1366_*` with their logs; converter `totalseg_to_speed.py` and `abdominal_acoustic_parameters.csv`.
- Abdomen figure `abdomenmultiscale.png`, commit `2de3791` (September 26, 2026), and its log `abdomenmultiscale_log.txt`; CT map figure `abdomen_acoustic.png`.

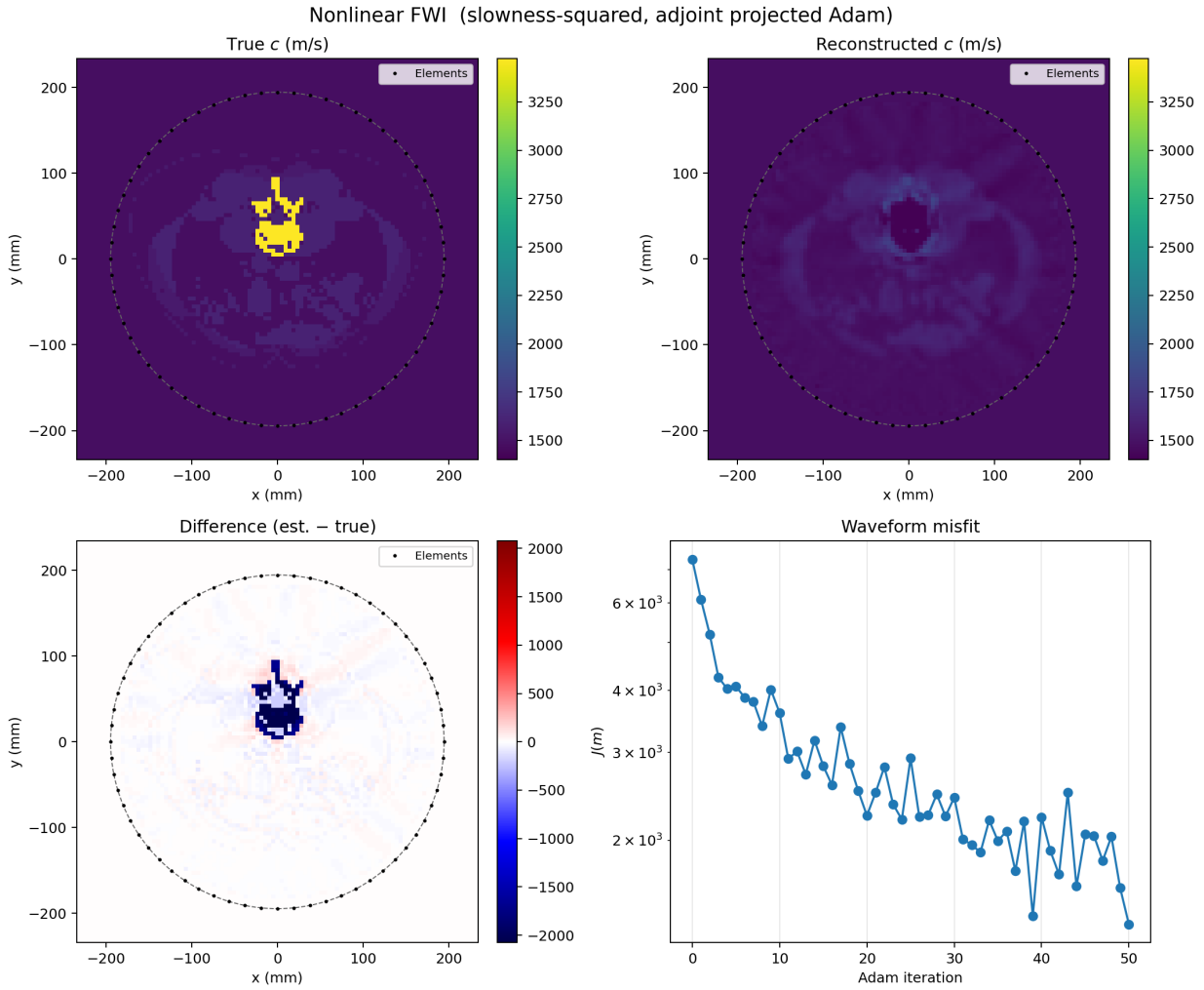


Figure 7: Coarse stage only (101^2, 25–75 kHz, 50 Adam iterations). Organ and muscle outlines are already visible; the vertebra region is driven down to the 1400 m/s lower bound.

- CT slice: `ct-abdomen-c4kc-kits-instance.dcm`, C4KC-KiTS collection, The Cancer Imaging Archive.
- Coarse-stage reproduction: run on October 27, 2026; gradient check: run on September 28, 2026 (both CPU, PyTorch 2.6).

References

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[3] T. D. Mast, "Empirical relationships between acoustic parameters in human soft tissues," *Acoustics Research Letters Online*, vol. 1, no. 2, pp. 37–42, 2000.

[4] J. Wasserthal *et al.*, "TotalSegmentator: robust segmentation of 104 anatomic structures in CT images," *Radiology: Artificial Intelligence*, vol. 5, no. 5, e230024, 2023.

[5] J. Wasserthal, "Small subset of TotalSegmentator Dataset," version 2.0.1, Zenodo, 2023, https://zenodo.org/records/10047263 (CC BY 4.0).

[6] M. O. Culjat, D. Goldenberg, P. Tewari, and R. S. Singh, "A review of tissue substitutes for ultrasound imaging," *Ultrasound in Medicine & Biology*, vol. 36, no. 6, pp. 861–873, 2010.